

Eren Dogan

Last updated: October 22, 2025

PhD Student in Computer Science and Engineering
Baskin School of Engineering
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EDUCATION

PhD in Computer Science and Engineering

GPA: 3.86/4.00

Advisor: Prof. Matthew Guthaus

University of California, Santa Cruz

September 2022 – June 2027 (expected)

BS in Computer Science

GPA: 4.00/4.00

Honors: Valedictorian, High Honor, Full-tuition scholarship

Ozyegin University

September 2017 – June 2022

RESEARCH

Publications

- M.1 **Eren Dogan**, Matthew R. Guthaus
“Effective Capacitance Modeling Using Graph Neural Networks.”
Manuscript on [arXiv](#).
- C.1 Bugra Onal, **Eren Dogan**, Muhammad Hadir Khan, and Matthew R. Guthaus
“GAT-Steiner: Rectilinear Steiner Minimal Tree Prediction.”
Paper at [43rd IEEE/ACM International Conference on Computer-Aided Design \(ICCAD 2024\)](#), Newark, NJ, USA, October 2024.
- P.1 **Eren Dogan**, H. Fatih Ugurdag, Hasan Unlu
“Deep Compression for PyTorch Model Deployment on Microcontrollers.”
Poster at [35th International Conference on VLSI Design and 21st International Conference on Embedded Systems \(VLSID 2022\)](#), Bengaluru, Karnataka, India, February 2022.
- W.1 **Eren Dogan**, H. Fatih Ugurdag, Matthew R. Guthaus
“OpenCache: An Open-Source OpenRAM Based Cache Generator.”
Paper at [4th Workshop on Open-Source EDA Technology \(WOSET 2021\)](#), Zoom, November 2021.

Contributions

- 2025- Contributor to [OpenROAD](#), an open-source RTL-to-GDS Flow.
- 2022-24 Contributor to [OpenRAM](#), an open-source static random access memory (SRAM) compiler.
- 2021 Developed [OpenCache](#), an open-source custom cache generator.

TEACHING EXPERIENCE

Teaching Assistant

University of California, Santa Cruz

Introduction to Analysis of Algorithms (CSE 102)

Spring 2025

Introduction to Data Structures and Algorithms (CSE 101)

Fall 2022, Spring 2024

Ozyegin University

Computer Architecture (CS 240)

Fall 2020

Digital Systems (EE 203)

Spring 2020

Computer Programming (CS 101)

Fall 2019

SERVICE

Reviewer

2025 *62nd ACM/IEEE Design Automation Conference (DAC 2025).*

2024 *32nd IFIP/IEEE International Conference on Very Large Scale Integration (VLSI-SoC 2024).*

2024 *6th ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD 2024).*

Volunteer

2025 *7th ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD 2025).*